SIT215 Computational Intelligence
Assignment 1: Problem Solving

Grading and weighting: (% total mark for unit): 40%

Submission Deadline: 8:00 pm AEST, 19 April 2026 (Week 6 Sunday)

Learning Objectives

This assignment focuses on solving real-world problems using computational intelligence (CI) techniques. It is designed to enable you to demonstrate your learning achievements in ULO1, ULO2, and ULO3. of this unit, including:

- Formulating real-world delivery challenges using formal computational intelligence representations.
- Exhibiting proficiency in selecting and applying appropriate search algorithm(s), model, and data structures to solve a constrained navigation problem.
- Developing, evaluating, and comparing alternative approaches using standard performance metrics.
- Communicating the design decisions, implementation logic, and evaluation outcomes clearly and professionally.

Assignment Overview - Problem to Solve

In this assignment, you will create a search-based navigation system for a truck delivery agent. Your tasks include modelling a delivery environment, accounting for truck-specific constraints, and designing search-based solutions that identify feasible and optimal delivery routes. This assignment requires you to integrate environment modelling, constraint-aware search, and evaluation of intelligence quality. You are not only expected to produce a working solution, but also to justify why your design choices lead to intelligent behaviour. This assessment task addresses the questions:

How can a constrained truck delivery problem be modelled and solved using search algorithms, and how can search intelligence be evaluated within a deterministic environment?

Your solution must demonstrate:

- environment modelling and abstraction
- PEAS formulation
- A* implementation and comparison with one uninformed search algorithm
- logical constraint validation
- design and evaluation of a new heuristic

These integrate concepts from Weeks 1 - 4. Note, the environment to model in this assignment is deterministic.

Submission

This is an individual assessment task. Please read this assignment tasksheet carefully. You **must** submit the following files to the relevant Assignment Dropbox folder on the Unit Site no later than the given deadline:

1) **The main report:** *[YourID]_assign1problemsolving_report.pdf***.** Your report must clearly present your solution, design decisions, and evaluation. Every submission must follow the standardised report structure, including *Cover Page, Problem Formulation, Modelling Approach, Search Strategies and Heuristic Design, Implementation Overview, Results and Evaluations, Conclusion, Acknowledgement (if any),* and *References sections.* Refer to Additional Guide for You to Complete this Assignment Effectively of this tasksheet for details.

2) **Source code: Jupyter Notebook (*[YourID].ipynb*) or Python files (*[YourID].py*).** All codes were developed or used as your complete solution in this assessment. Your code must be in appropriate documentation with sufficient comments, explaining the key logic of your implementation.

3) **README file:** *[YourID]_README.txt* **or** *[YourID]_README.md* with clear execution instructions, specifying any required libraries or dependencies.
   - Note: If your code solution includes the usage of any open-source libraries, you need to provide those libraries or make them accessible to the assessor (e.g., by providing a link to a download site and instructions on how to install and use the library in your solution).

Late Penalties

In accordance with Faculty assessment policies, late submissions to the submission folder will incur a penalty based on Deakin's Penalties for late submission and Due dates policies, https://www.deakin.edu.au/students/study-support/assessments-and-examinations/assessments

Your Tasks

An Assignment 1 submission must present a complete and coherent problem-solving solution. All Tasks 1–4 are compulsory and will be evaluated together as an integrated computational intelligence solution.

Task 1: Environment modelling and problem formulation

Your modelling should clearly demonstrate how a real-world delivery problem is abstracted into a state space suitable for computational intelligence techniques.

1.1) Create an environment to address the given problem using an appropriate representation.

- Select a real environment (e.g., campus, city block, logistics area) and model it with 20 – 23 landmarks.
- Your model must:
  - include around 35-38 meaningful path segments
  - clearly define nodes and paths, and associated path costs
  - consider at least two realistic constraints that influence path feasibility – one static (time/condition-independent) and one dynamic (time/condition-dependent), as a part of your problem formulation (description and justifications are needed in your report), such as:
    - minimum road width or turning radius,
    - restricted access zones,
    - any moving obstacle on the road

    but not limited to the above-mentioned factors

Notes:

- The guide with around 20 landmarks and 35 path segments in your graph is a reasonable configuration for you to implement A* effectively in Task 2.
- You can gather environmental information using physical surveys, approximate sketches, online research, virtual street views, or collaboration with local disability organisations. You can also employ mapping tools like Google Maps (Google My Maps: https://support.google.com/mymaps/), OpenStreetMap (https://www.openstreetmap.org/). If exact measurements are unavailable, you may use reasonable figurative estimations to represent environmental features clearly.

1.2) In your report – sections: *Problem Formulation*, and *Modelling Approach*, include the following (but not limited to)

- Describe your defined problem, goal, and environment. Justify the features and landmarks included in your map and modelled environment, respectively.
- Define the Performance-Environment-Actuator-Sensor (PEAS) framework aligned with your defined problem and use it to justify your modelling choice.
- Explain how the real environment was abstracted into your state space and what constraint(s) are removed in the relaxed version.

Task 2: Search-based approaches implementation

You are required to implement and compare search-based algorithms for your relaxed problem (only your defined static constraint(s) are enforced). Your code must be clearly structured (e.g. state, successor function, etc) and annotated to show how the search process operates. Your deliverables must include:

2.1) Implementation of A* as the benchmark algorithm, Manhattan discussion as the baseline heuristic.

2.2) Implementation of one uninformed search algorithm of your choice (such as Breadth-first search, Depth-first search, or Uniform Cost Search), you will use this algorithm for comparing with the benchmark A*.

2.3) In your report – *sections*: *Modelling Approach*, *Implementation Overview*, and *Search Strategies and Heuristic Design*, explain the following (but not limited to)

- why your selected uninformed-search algorithm is suitable for comparison with A* benchmark
- explain why Manhattan distance is appropriate for your environment
- demonstrate your search implementation by showing proper data structure and the key components of node expansion logic, supported by annotated code in your report. (Via a citation by line numbers of your provided source [code, file] *or* screenshots of code snippet embedded in your report, for effective assessment.)
  - Note: Code logic is assessed through annotations, explanations, and referenced snippets in the report. Code must be sufficiently documented so that its behaviour can be understood without step-by-step debugging.

Task 3: Success metrics and evaluation

You must evaluate both search algorithms using at least three start–goal pairs on your relaxed problem, and compare their behaviour using standard performance metrics. Present results in structured tables or visualisations where appropriate.

3.1) Your evaluation must analyse completeness, optimality, and efficiency.

3.2) In your report – *sections*: *Results and Evaluations*, include the following (but not limited to)

- Compare algorithm performance using appropriate metrics
- Explain why certain algorithms behave better or worse in your modelled environment
- Interpret what the results indicate about intelligence quality

Task 4: New heuristic design and performance

4.1) Design and implement a new heuristic for A*.

Note: The new heuristic does not need to be complex, but it must differ meaningfully from the baseline heuristic and be justified using deterministic knowledge of the environment.

4.2) In your report – *sections*: *Implementation Overview*, *Search Strategies and Heuristic Design*, and *Results and Evaluations*, include the following (but not limited to)

- Explain your new heuristic's logic and how it is constructed.
- Justify whether it is admissible, and elaborate its impact using the standard search performance metrics.
- Compare it with Manhattan distance (baseline heuristic from Task 2), analyse which one is stronger and why within your modelled problem and environment.

Marking

Your submission will be evaluated based on criteria that reflect the depth of your problem-solving skills, technical proficiency, theoretical knowledge, and communication effectiveness against the quality of your work. A numeric mark (out of a total mark of 40) will be determined by how well your work aligns with the grading criteria and its standard as outlined in the rubric available on the Unit Site under Resources>Assessment.

Additional Guide for You to Complete this Assignment Effectively

Preparatory Learning Activities

To complete this assessment task, you will need to first develop an understanding of a range of topics covered in this unit in weeks 1 to 4. Given the assessment deadline, this may require you to learn these topics independently to understand the scope as soon as possible in earlier weeks. The topics that you will need to be familiar with are:

- Agents and Environment
- Path Search Strategies
- Heuristic
- Solution Quality, Evaluation, and Logical Reasoning

Understanding these areas and how they inter-relate will make it far easier to understand the learning materials, and explain your investigations and outcomes in this assignment. Our advice is that you use this assignment as a basis for the study of these underlying areas to assist in integrating the knowledge covered in this unit into a meaningful 'whole', which supports completing this assessment task.

Report Structure

You may reorder the following sections where appropriate to reflect the logical progression of your work.

<table>
<tr><th>Sections</th><th>Guide for you</th></tr>
<tr><td>Cover Page</td><td>Must provide checklist below to ensure clarity in your submission. Tick (✓) the boxes next to the tasks you completed to help you verify your work before submission. Your submission will be assessed based on the tasks you mark as completed. Ensure each ticked task is fully supported with corresponding sections in both your report and code.
<table>
<tr><th>Tasks</th><th>Completed (✓)</th><th>Notes (Optional)</th></tr>
<tr><td>Task 1: Environment modelling and problem formulation</td><td>☐</td><td></td></tr>
<tr><td>Task 2: Search-based agent implementation and comparison
• A*
• Selected uninformed algorithm</td><td>
☐
☐</td><td></td></tr>
<tr><td>Task 3: Success metrics and evaluation</td><td>☐</td><td></td></tr>
<tr><td>Task 4: New heuristic design and performance</td><td>☐</td><td></td></tr>
</table></td></tr>
<tr><td>Problem Formulation</td><td>This establishes the problem definition and context, and constraints in your own words. Show your map with clear landmarks, and graph, etc in this section.</td></tr>
<tr><td>Modelling Approach</td><td>Explains your environment model and agent, using PEAS framework. Explain the relaxed version of your problem.
Hints: Focus on CI concepts across aspects like environment, search, and agent design that impact your approach. Use appropriate terminology and relevant examples from class discussions or practical exercises to justify your choices. You can further cite with your class discussions, peer discussion on unit site forum/Teams to strengthen your explanation.</td></tr>
<tr><td>Search Strategies and Heuristic Design</td><td>Briefly introduce the benchmark A*. Explain and justify your choice of uninformed search algorithm for comparison with the benchmark. Analyse the heuristic function(s) used, and justify their properties and relevance within the modelled environment. Focus on reasoning and justification rather than restating textbook definitions.</td></tr>
<tr><td>Implementation Overview</td><td>Provide a clear high-level narrative explaining the key components of your solution (e.g. state, successor, cost tracking, search logic, but not limited to). Ensure your elaboration aligns with (but does not duplicate) your code comments, against your defined problem goals. References to code snippets (by indicating the line numbers of your submitted code program) or paste the screenshot of your code snippets (must show visible line numbers as well) directly in your report.</td></tr>
<tr><td>Results and Evaluations</td><td>Interpret and evaluate the outcomes of your working solution in terms of intelligence quality. Use appropriate visual aids (e.g., graphs, charts, tables, or figures) where relevant to support your analysis. Discuss the significance of these results and what they indicate</td></tr>
</table>

	about the performance and effectiveness of your solution, ensuring that all feasible test cases introduced are clearly compared.
Conclusion	Summarise the key findings of your work, linking them explicitly to your implementation and evaluation results. Conclude with a brief logical validation discussing whether solutions remain reachable under your defined constraints and whether feasible paths satisfy the intended delivery requirements.
Acknowledgement	Record any external assistance, if applicable.
References	List all citations and resources used, using an appropriate academic referencing style.

Expected Report Length

Your report should be concise yet comprehensive, focusing on the key components of your solution while maintaining depth and coherence. Where relevant, use appropriate visuals (e.g., maps, graphs, tables, figures, or flowcharts) to support your analysis. All sources must be cited and referenced using a consistent academic referencing style. If completed correctly, *as a guide*, the expected length is approximately 8 to 12 single column A4 pages, using font size of 11 with single line spacing, covering all essential sections as the outlined.

Report Structure:

- Cover Page: include only the required checklist table, along with your name and student ID
- Problem Formulation & Modelling Approach: 1-2 page(s)
- Search Strategies and Heuristic Design: 2-3 pages
- Implementation Overview: 2-3 pages with annotated code snippets
- Results and Evaluations: 2-3 pages, including proper and relevant visuals
- Conclusion: 1 page
- Acknowledgements (if any) and References: no strict page limit

Getting Help and Support

Students are encouraged to collaborate by discussing tasks and assisting each other in understanding concepts, models, and algorithms. Engaging in class sessions (Lecture, Prac) and participating in the unit site discussion forum or Teams support channel can strengthen your understanding and contribute to peer learning. However, this is an individual assessment, and all development work and report writing must be your own. If using external references, ensure they are properly cited. Collusion and plagiarism will result in severe academic penalties. If you need clarification on completing this task, seek guidance from the teaching team.

Guidance Support

The teaching team is available to assist you with this task. If you need help understanding the material or requirements, seek support during class time or post your questions in the Discussion Forum on the Unit Site, where responses can also benefit your peers.

Report Writing Help

While the teaching team can offer guidance on report writing, ensure you review the instructions in this task sheet carefully. Additional support is available through the Writing Mentors program, which provides assistance with academic writing, including report preparation. For more information, visit http://www.deakin.edu.au/students/studying/study-support/writing-mentors.

Feedback

Students are encouraged to actively consolidate their design inspiration from peer discussion in class, and seek formative feedback from teaching staff on their work completed before the submission deadline to ensure they are on track with this task. Formative feedback may be obtained during weekly scheduled practical classes upon request. Talk to us, and we'll support you! Besides, in 15 business days after the assignment submission due date, marking feedback will be provided via the unit site.

End of Assignment 1 Tasksheet.